

M2 — Critique of another group’s M1 (Milestone 2 slide deck)

*Model submission. We are **Group B**. We were assigned **Group A’s M1** on Kueh et al., BMJ Open 2026;16:e113719 — the deck reproduced in `../M1_replication/`. Read that deck first; this one is the review of it.*

Written with Group A’s deck and repository in front of us and nothing else. We do not know what they will do with any of this at M3.

Slide 1: What we reviewed, and where we land

- **Reviewed:** Group A’s M1 deck (8 slides) and their repository zip.
- **Their headline:** the paper reproduces; they hold **two estimands apart** throughout — the replication contrast (IV vs I) and their locked scientific contrast (IV vs III) — and they name **estimand–adjustment-set alignment** as the dominant threat.
- **Our overall view:** a disciplined replication. The four-cell diagnostic on Slide 4, computed on one fixed row set, is better practice than most published sensitivity analyses, and their missing-data slide reports a null result honestly instead of dressing it up.
- **We have two objections about evidence, and we nominate a different dominant threat.** We are not asking them to start over.

Speaker notes: Open with the verdict. Note what we are *not* doing: we are not manufacturing a missingness objection, because Slide 6 already shows the locked estimate moving by a hundredth at most across complete case, both extreme assignments and MI. Re-litigating a question the authors already answered would be padding, and reviewers who pad get read less carefully on the points that matter.

Slide 2: Did their code run?

- **Yes**, end to end from raw data. We regenerated the sample funnel, the group sizes, the death count and both headline contrasts, and they match the deck.
- **The design question is answered in the code, and answered correctly.** They build the survey design on the full frame and `subset()` it to the MASLD-eligible domain, rather than filtering rows before `svydesign()`. We looked specifically because this is the most common NHANES error in the applied literature; they did not make it.
- **Their numbers are not typed into the deck.** Every figure is substituted at build time from emitted result objects, so none can be mistyped. That prevents retyping errors, not stale results: if the data change and the result objects are not re-emitted, the deck still shows the old values. We re-ran the increments before comparing, so the match above is against freshly emitted results.
- **One gap:** their README does not list every package the pipeline loads, so a clean install stops twice before completing.

Speaker notes: “It ran” is not a finding. Say what was regenerated and what was checked deliberately. The design-subsetting check is worth naming explicitly *because* it came back clean — a reviewer who only reports faults gives the authors no information about what has been verified.

Slide 3: Objection 1 — the published table cannot carry the weight placed on it

- Slide 4 supports the alignment argument partly with the paper’s own Table 2: published Model 1 gives Group III **3.175** and Group IV **3.155** against Group I, an implied IV-vs-III ratio of about **0.99**.
- Their own detail note concedes the interval for that implied ratio **cannot be recovered**, because the covariance of the two published coefficients is not reported.

- So the deck cites a point estimate with **no uncertainty statement** as corroboration for a claim about compatibility with no difference. Two point estimates can imply a ratio near 1.00 whether that ratio's own interval is narrow or wide, and the interval depends on a covariance that is not reported.
- **What we are asking for:** keep the observation, demote the claim. “The published table is consistent with our estimate” is defensible; “the paper’s own Table 2 points the same way” reads as independent confirmation, and it is not independent — it is the same data, without an interval.

Speaker notes: This is a reasoning objection, not a coding one, so running the pipeline would never surface it. We are not saying the implied ratio is wrong. We are saying it is being used as evidence when it is at best consistency, and the deck’s own detail note already knows this.

Slide 4: Objection 2 — the TyG check is listed as a remedy but is not one

- Slide 8 lists “a design-aware re-analysis of the TyG sensitivity on the locked contrast” under *what would change our mind*.
- The authors’ TyG sensitivity reports about **3.1** for both Group III and Group IV against Group I. That is the **replication contrast**, unweighted — precisely the two properties the deck spends Slide 4 arguing are the wrong ones for the locked question.
- Re-running it design-aware would not fix that. The contrast would still be against Group I, so it cannot adjudicate IV versus III whatever weights are applied.
- **What we are asking for:** either drop it from the list, or state what a TyG re-analysis would actually have to estimate — the IV-vs-III contrast under a TyG-based exposure definition — which is a different analysis from the one the authors ran.

Speaker notes: This is the point we expect most pushback on, and it is fair pushback: they might reply that a design-aware TyG check speaks to exposure misclassification rather than to the contrast. If so, say that on the slide, because as written the item reads as though it would resolve the dominant threat, and it would not.

Slide 5: We agree alignment matters — we do not think it dominates

- We accept that the target quantity is unstated and that this is consequential. We are **not convinced the evidence presented resolves it**, for the two reasons above.
- But an unstated estimand is a **reporting** gap, and it is repairable by the authors in a sentence. We think a **structural** limitation outranks it.
- **Our nomination: selection into the analytic cohort.** The cohort is reachable only through the fasting subsample — of roughly **36,580** adults, **14,962** were sampled to fast and about **14,170** of those have a computable FLI, and everything downstream is conditioned on that. The deck does not show that the people who end up with a computable FLI represent the population the conclusion is addressed to.
- That is not fixable by relabelling the estimand. It bounds what any contrast estimated here can be said to be about, including the locked one.
- **Why this outranks alignment:** naming the target quantity makes the estimate interpretable; it does not make it transportable. The selection is upstream of every contrast in the deck.

Speaker notes: A reviewer is entitled to disagree about which threat dominates — that is reviewing rather than auditing the authors’ self-diagnosis. Be explicit that we are not saying they were wrong to be worried about alignment; we are saying they ranked a repairable reporting problem above an unrepairable structural one.

Slide 6: Recommendations, in priority order

1. **Demote the Table 2 corroboration** to a consistency observation, and say why no interval is available (Slide 3).
2. **Restate or drop the TyG item** on the “what would change our mind” list, since as written it cannot settle the locked contrast (Slide 4).
3. **Address selection explicitly** — state the population the locked contrast is about, given that entry requires a computable FLI, and say what would be needed to speak beyond it (Slide 5).

An answer to their open question. Their Slide 8 asks reviewers whether “the estimand is unstated” is a finding or a refusal to commit to a causal model themselves. Our answer: it is a finding, but a weaker one than the deck treats it as. Declining to name a target quantity is a real limitation in the paper. It becomes a *dominant* threat only if naming it would change what the analysis can support — and here, we think selection would still be binding afterwards.

Speaker notes: Three actionable items, ordered, none of which requires them to start over. Closing on their own question is deliberate: they invited the challenge, and a review that ignores an explicit invitation has not engaged. We expect them to push back on the ranking in Slide 5, and that is the argument worth having at M3.

Slide 7: Cover note

Contribution. AB: ran their pipeline, reproduction log, Slides 1–2. CD: the two evidence objections, Slides 3–4. EF: the dominant-threat disagreement and the recommendations, Slides 5–6, deck assembly.

External feedback. None received.

AI Use Statement. Tool and version: Claude (Opus 5), in a browser. Where we used it: resolving two package-installation failures on our own machines while setting up to run Group A’s repository, and tightening the wording of Slides 3 and 5 after we had written them. What we did with the output: we rewrote both slides from the tightened draft and checked every claim back against Group A’s deck and their scripts. **No part of Group A’s deck, repository, output or results was pasted, uploaded, screenshotted or retyped into any AI tool;** where an error message quoted their code, we described the problem in our own words instead. The critique, including the ranking on Slide 5, is ours.

Speaker notes: M2 submits slides only, so there is no `COVER-NOTE.md` to file — this slide is the whole cover note. The two hard limits at M2 are worth stating on the slide itself rather than assuming the reader takes them on trust.